

JUNIT::

EXERCISE 1:

Calculator.java:

```
package exercise1;

public class Calculator {

    public int add(int a, int b) {

        return a + b;

    }

}
```

CalculatorTest.java:

```
package exercise1;

import org.junit.jupiter.api.Test;

import static org.junit.jupiter.api.Assertions.*;

public class CalculatorTest {

    @Test

    void testAddition() {

        Calculator calc = new Calculator();

        assertEquals(5, calc.add(2, 3));

    }

}
```

EXERCISE 4:

Calculator.java:

```
package exercise1;

public class Calculator {

    public int add(int a, int b) {

        return a + b;

    }

}
```

```
}  
}
```

CalculatorTest.java:

```
package exercise4;  
  
import org.junit.jupiter.api.AfterEach;  
import org.junit.jupiter.api.BeforeEach;  
import org.junit.jupiter.api.Test;  
  
import static org.junit.jupiter.api.Assertions.assertEquals;  
  
public class CalculatorTest {  
    private Calculator calculator;  
  
    @BeforeEach  
    void setUp() {  
        calculator = new Calculator();  
        System.out.println("Setup: Calculator object created");  
    }  
  
    @Test  
    void testAddition() {  
        int a = 10;  
        int b = 20;  
        int result = calculator.add(a, b);  
        assertEquals(30, result);  
    }  
  
    @AfterEach  
    void tearDown() {  
        System.out.println("Teardown: Test completed");  
    }  
}
```

MOCKITO::

EXERCISE 1:

ExternalApi.java

```
package mockitoexercise1;

public interface ExternalApi {

    String getData();

}
```

MyService.java

```
package mockitoexercise1;

public class MyService {

    private ExternalApi api;

    public MyService(ExternalApi api) {

        this.api = api;

    }

    public String fetchData() {

        return api.getData();

    }

}
```

MyServiceTest.java:

```
package mockitoexercise1;

import org.junit.jupiter.api.Test;

import static org.junit.jupiter.api.Assertions.assertEquals;

import static org.mockito.Mockito.*;

public class MyServiceTest {

    @Test

    void testExternalApi() {

        ExternalApi mockApi = mock(ExternalApi.class);

        when(mockApi.getData()).thenReturn("Mock Data");

    }

}
```

```

    MyService service = new MyService(mockApi);

    String result = service.fetchData();

    assertEquals("Mock Data", result);
}
}

```

pom.xml:

```

<?xml version="1.0" encoding="UTF-8"?>

<project xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/xsd/maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <groupId>or.example</groupId>

    <artifactId>TDD</artifactId>

    <version>1.0-SNAPSHOT</version>


    <properties>

        <maven.compiler.source>26</maven.compiler.source>

        <maven.compiler.target>26</maven.compiler.target>

        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>

    </properties>


    <dependencies>

        <dependency>

            <groupId>org.junit.jupiter</groupId>

            <artifactId>junit-jupiter</artifactId>

            <version>5.10.2</version>

            <scope>test</scope>

```

</dependency>

<dependency>

<groupId>org.mockito</groupId>

<artifactId>mockito-core</artifactId>

<version>5.12.0</version>

<scope>test</scope>

</dependency>

</dependencies>

</project>